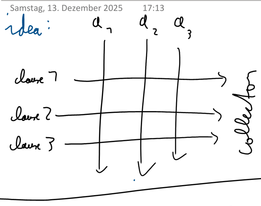
Minesweeper - Component Definition

Samstag, 13. Dezember 2025 17:13

similar to MC Vaten

idea:



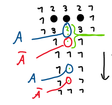
CNF formula satisfiable
⇕
There exists a layout of bombs such that the minesweeper board is valid

Notation:

numbers have the same function they have in Minesweeper
all completely empty space is a 0

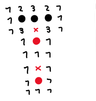
- ● ... predefined Bomb
- ● ... input-dependent Bomb
- × ... input-dependent safe field (no Bomb)
- ? ... value of variable: true → × , false → ●
- · ... field without value, can become × or ● (dependent on input)

Input stream for a variable:



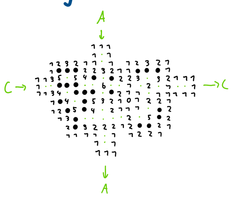
information gets carried via stream

A = true

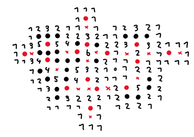


A = false

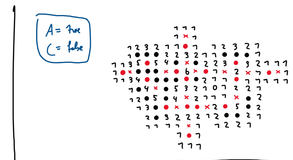
Crossing: clause does not use variable, stream do not change



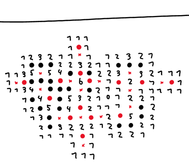
A false
C false



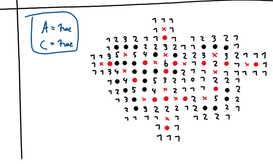
A = true
C = false



A false
C true



A = true
C = true



Clause stream

checks variable specified in a clause is 1 if at least one literal satisfied the clause

start with 0:

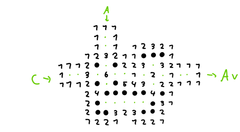
Note: we are „short" of one point
but by choosing a wrong option
we get an error, so crossing is valid

(this only works, but 7 would be barely enough here)

↓ wrong bomb field at 6

6 for 7 bomb!
NOT VALID

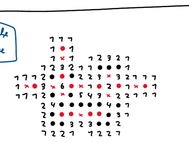
OR: clause is true if clause was already true or if current literal is true



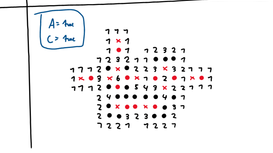
A = false
C = false

A = true
C = false

A = false
C = true

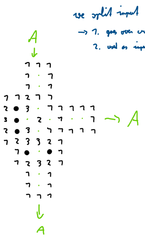


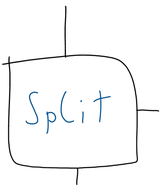
A = true
C = true



Splitter: Because OR does not keep variable input as secondary output,

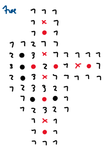
we split input stream OR to two equal stream
→ 1. goes over crossing to continue input towards
2. used as input for OR





A = false

A = true



Collector: needs to receive true

otherwise clause is false → not satisfying assignment → not valid minesweeper configuration

C = true

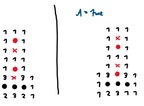
valid ✓

C = false

→ don't have Bomb
Board invalid! ✗

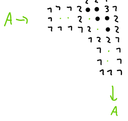
End: ends input stream

A = false



A = true

Turn stream



Offset stream (horizontally)

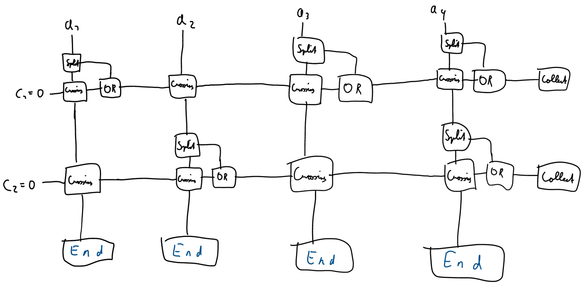
exception: no offset

Offset stream (vertically)

needed for clause
to read negation of variable
(when it uses negated literal)

Example

$$\underbrace{(a_1 \vee \overline{a_3} \vee a_4)}_{C_1} \wedge \underbrace{(a_2 \vee \overline{a_4})}_{C_2}$$



+ Offsets where they are needed